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# Compact support is not enough: sparse-plus-spectral kernels for exact Gaussian processes on molecular torsion spaces

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## Abstract

Exact Gaussian processes can be scaled to millions of points by giving the kernel compact support, so that the covariance matrix is genuinely sparse rather than merely approximated. We ask what happens when this idea meets a domain whose input space is a torus and whose dimension is six: the relaxed backbone torsional potential-energy surface of a capped tripeptide, a standard object in force-field parameterisation. Fitting an exact GP to 500,000 conformations on one 32-core node, we find that compact support alone fails, and fails for a reason that is intrinsic rather than incidental. Any support radius that keeps the covariance solvable in memory leaves, in six dimensions, a matrix whose off-diagonal mass is concentrated on a handful of near neighbours; the resulting model is barely better than a constant plus noise. Making the kernel periodic repairs a real artefact — a ridge of inflated posterior variance along every  $\pm\pi$  seam — but does not repair the starvation. We therefore construct the *torsional ANOVA–Wendland* kernel, which adds to a periodic, non-stationary, *screened product* Wendland term an exactly low-rank ANOVA–Fourier term whose feature map is the many-body torsional expansion of a molecular-mechanics force field. The covariance is exactly sparse-plus-low-rank, so mat-vects, the linear solve and the log-determinant all remain exact. On a frozen 10,000-point test set the construction reaches CRPS 2.383 against 2.563 for the best compactly supported kernel of the original framework (Eq. (9)) and 2.562 for a tuned SVGP, with near-nominal 95% credible-interval coverage 0.940 that the strongest approximate baseline (SVGP, 0.709) badly misses. Two findings come out of the fit itself. The learned amplitudes show that the surface is strongly *non-additive* — two-body torsion–torsion terms carry 60.7% of the prior variance against 17.2% for the one-body terms — so a separable model is structurally inadequate, and the dominant couplings are between same-type backbone torsions on different residues rather than the intra-residue  $(\phi, \psi)$  pair we had pre-registered as our expectation. And a direct measurement shows that 4.0% of conformation space sits on a genuine discontinuity of the adiabatic surface, a property of relaxed torsional scans that sets a noise floor of 2.03 against a total spread of 5.65 and bounds what any regression of such data can achieve.

## 1 Introduction

The energy of a peptide as a function of its backbone torsions is one of the most heavily used objects in molecular modelling. Ramachandran’s map of the allowed  $(\phi, \psi)$  regions [Ramachandran et al., 1963] organises protein structure; modern force fields carry an explicit two-dimensional torsion–torsion correction term, the CMAP, because a sum of one-dimensional torsion terms is not accurate enough [MacKerell et al., 2004]; and force-field parameterisation proceeds by fitting to relaxed torsional scans [Qiu et al., 2021]. What a modeller wants from such a surface is not only a prediction but a calibrated statement of where the surface is known and where it is not, because that is what

decides where to spend the next expensive quantum calculation. This is exactly how Gaussian processes are already used for one-dimensional hindered-rotor scans, where the posterior variance chooses the next fidelity or grid point [Fleck et al., 2024]; the full backbone surface, however, couples several torsions at once, and there scale becomes the obstacle.

Gaussian process regression is the natural tool [Rasmussen and Williams, 2006], and it has a long record in this field [Bartók et al., 2010, Chmiela et al., 2017, Deringer et al., 2021]. Its obstacle is scale: an exact GP on  $N$  points needs  $\mathcal{O}(N^2)$  memory and  $\mathcal{O}(N^3)$  time. The dominant response has been approximation — inducing points [Titsias, 2009, Hensman et al., 2013], nearest-neighbour variational families [Wu et al., 2022], structured kernel interpolation [Wilson and Nickisch, 2015], Vecchia factorisations [Vecchia, 1988, Katzfuss and Guinness, 2021] — each of which replaces the covariance, and therefore the uncertainty, by something else.

A different response is to observe that a covariance matrix is dense only because ordinary kernels declare every pair of points correlated. A compactly supported kernel makes the matrix genuinely sparse, and the GP stays exact. This is classical as *covariance tapering* [Furrer et al., 2006, Kaufman et al., 2008]; it becomes a general-purpose scaling strategy once the compactly supported kernel is also flexible and non-stationary, which is the route taken by Risser et al. [2025] and pushed to ten million points by Noack et al. [2025]. Those demonstrations are all low-dimensional and Euclidean — one-dimensional synthetic data, topography, three-dimensional temperature fields — with one eight-dimensional exception on which the method is reported as merely competitive.

This paper asks what the sparsity-discovery strategy does on a domain that is neither low-dimensional nor Euclidean, and finds that the answer is instructive. Our contributions are as follows.

**A quantitative failure mode of compact support in moderate dimension.** On the six-torus with  $N = 500,000$ , a sparsity budget that fits a single node forces a per-coordinate support half-width of order  $50^\circ$ . In six dimensions the volume of that box is a  $10^{-4}$  fraction of the domain, and the Wendland profile decays so fast across it that the median surviving off-diagonal entry is nearly 3.1 orders of magnitude below the diagonal, with 52% of them below one part in a thousand (Fig. 2c). The covariance is numerically almost diagonal, and a GP built on it is barely distinguishable from a constant plus noise. We quantify this for the kernels of Noack et al. [2025] run on our own domain under an identical budget and solver.

**A kernel whose global part is exactly low-rank and physically motivated.** We construct the torsional ANOVA–Wendland (TAW) kernel,  $k = k_{\text{AF}} + k_{\text{TW}}$ . The local term  $k_{\text{TW}}$  is a periodic, non-stationary, *screened product* Wendland on  $\mathbb{T}^d$ : it is built from per-circle chordal distances, so it respects the geometry; its support is a box rather than a ball; and its per-circle length scales may individually grow past the diameter of their circle, at which point that coordinate stops screening — so the hyperparameters choose *which* torsions must be resolved locally. The global term  $k_{\text{AF}}$  is an ANOVA expansion of periodic correlations truncated at a few harmonics, whose feature map is precisely the functional form of a molecular-mechanics torsional energy: a constant, one-body torsion terms and two-body torsion–torsion cross terms. It is globally supported but of exactly finite rank  $R = 661$ , so the covariance is  $K = S + UU^\top + \nu I$  with  $S$  sparse — still exact, with no inducing points and no interpolation, and with mat-vecs, solves and log-determinants that cost what the sparse part costs plus  $\mathcal{O}(NR)$ .

**Two findings about the domain, read off the fitted model.** The learned amplitudes decompose the surface by interaction order, and the decomposition is lopsided: the two-body torsion–torsion terms carry 60.7% of the prior variance against 17.2% for the one-body terms and 19.7% for the compactly supported remainder. The surface is therefore strongly non-additive, which is both the reason a purely local product kernel cannot model it and a statement about the molecule — and the coupling is concentrated not on the intra-residue  $(\phi_i, \psi_i)$  pair we pre-registered as our expectation but on same-type backbone torsions across residues (68% of the two-body variance), a signature of cooperative backbone relaxation that we report and dissect honestly against that expectation. And a direct measurement of the surface shows that 4.0% of conformations sit on a discontinuity of the adiabatic surface: displacing all six torsions by 0.01 rad changes the energy by more than three compressed kcal/mol, because the constrained minimisation of the other 36 degrees of freedom changes basin. This sets a noise floor of 2.03 against a total spread of 5.65, explains why point metrics saturate, and is exactly the kind of statement a calibrated exact posterior can make and a variational one cannot.

## 2 Methodology

### 2.1 The regression problem

We write  $\theta = (\phi_1, \psi_1, \phi_2, \psi_2, \phi_3, \psi_3) \in \mathbb{T}^6 = (S^1)^6$  for the six backbone dihedrals of the capped tripeptide Ace-(Ala)<sub>3</sub>-NMe (42 atoms) and  $E(\theta)$  for the relaxed MMFF94s energy [Halgren, 1996, 1999]: the six torsions are held at  $\theta$  by stiff restraints while all other degrees of freedom are minimised, and the label is the *unrestrained* energy at the relaxed geometry. Because a uniform draw on  $\mathbb{T}^6$  is usually a strained conformation — the median relative energy is 46.0 kcal/mol and the 99th percentile is 377 kcal/mol — we regress the monotone compression

$$y = E_0 \log\left(1 + \frac{E - E_{\text{ref}}}{E_0}\right), \quad E_0 = 10 \text{ kcal/mol}, \quad (1)$$

with  $E_{\text{ref}} = -0.0106$  kcal/mol the global MMFF94s minimum. The map is strictly increasing, so credible intervals and hence the coverage curve are identical in  $E$  and in  $y$ ; we also report back-transformed errors in kcal/mol on the thermally accessible window.

### 2.2 What compact support costs in six dimensions

Let the kernel have box support of angular half-width  $h$  in each of  $d$  coordinates. For data uniform on  $\mathbb{T}^d$  the fraction of non-zero covariance entries is  $(h/\pi)^d$ , so a budget of  $\bar{m}$  non-zeros per row at sample size  $N$  fixes

$$h = \pi (\bar{m}/N)^{1/d}. \quad (2)$$

At  $N = 500,000$ ,  $\bar{m} = 300$  and  $d = 6$  this is  $h = 50^\circ$ : the support is comparable to the width of a Ramachandran basin, so one cannot make it smaller without emptying it, and cannot make it larger without leaving the memory of the node. The difficulty is not the radius but what happens inside it. The  $\bar{m}$  neighbours are spread through a six-dimensional box, so a neighbour at rank  $j$  sits at relative radius  $\approx (j/\bar{m})^{1/d}$ , and the Wendland profile  $W$  of Wendland [1995] — with the specific polynomial  $\varphi_{3,3}$  used by Noack et al. [2025] — has already fallen to  $W \approx 10^{-2}$  at half its support. The consequence, which we confirm empirically in Sec. 5.1, is that the sparse matrix is numerically close to diagonal. It is a genuine dimensional effect: the same construction in  $d \leq 3$ , where Noack et al. [2025] demonstrate it, gives  $h$  small enough that many neighbours sit deep inside the support. That local kernels forfeit their locality as the dimension of a molecular energy-fitting problem grows has been noted before for Gaussian-type kernels [Manzhos and Ihara, 2023]; what is specific to the exact-GP-at-scale setting is that the sparsity budget makes the loss unavoidable — the support cannot be shrunk without emptying it — so the remedy cannot be a better local kernel.

Two consequences shape the design below. First, among the Wendland family we use the *lowest* order compatible with the smoothness we need,  $\varphi_{3,1}(r) = (1 - r)_+^4 (4r + 1)$ , which retains far more mass near the edge of its support; this choice was made on a held-out validation split before any test-set evaluation and is applied identically to every kernel we compare (Table 3). Second, and more fundamentally, a purely local kernel cannot use the fact that a peptide’s torsional energy is nearly additive across its coupling graph, because additivity is a global statement.

### 2.3 A screened non-stationary Wendland kernel on the torus

For  $\theta, \theta' \in \mathbb{T}^d$  let  $c_k = 2|\sin \frac{1}{2}(\theta_k - \theta'_k)|$  be the chordal distance on circle  $k$  — the Euclidean distance between  $(\cos \theta_k, \sin \theta_k)$  and  $(\cos \theta'_k, \sin \theta'_k)$  in  $\mathbb{R}^2$ . With a periodic log-length-scale field  $\ell_k(\theta) = \bar{\ell}_k \exp\{\sum_{m=1}^2 a_{km} \cos m\theta_k + b_{km} \sin m\theta_k\}$  we define

$$k_{\text{TW}}(\theta, \theta') = \sigma_L^2 \prod_{k=1}^d \underbrace{\frac{2 \ell_k(\theta) \ell_k(\theta')}{\ell_k(\theta)^2 + \ell_k(\theta')^2}}_{p_k} W\left(\frac{c_k}{s_k}\right), \quad s_k = \sqrt{\frac{1}{2}(\ell_k(\theta)^2 + \ell_k(\theta')^2)}. \quad (3)$$

Each factor is the Paciorek–Schervish construction [Paciorek and Schervish, 2006] applied in the  $\mathbb{R}^2$  embedding of circle  $k$  with isotropic  $\Sigma = \ell_k^2 I_2$ , hence a valid kernel on  $S^1$ ; a Schur product of valid kernels is valid, so (3) is a valid kernel on  $\mathbb{T}^d$ . Three differences from Eqs. (3) and (6) of Noack et al. [2025] matter. It is periodic, so no artificial seam is introduced. Its support is a *box*

$\{c_k < s_k \forall k\}$  rather than a ball, so the sparsity pattern is a product of per-coordinate conditions and the anisotropy is exact rather than approximate. And because every factor lives in  $\mathbb{R}^2$ , the  $d \leq 3$  validity range of the Wendland polynomial is respected — a Euclidean Wendland applied in  $\mathbb{R}^6$ , as the original framework does on its eight-dimensional benchmark, is outside the range in which positive definiteness is guaranteed.

The construction is *screened* in the following sense. Support in coordinate  $k$  requires  $c_k < s_k$ , and  $c_k \leq 2$  always; if  $\bar{\ell}_k$  grows past 2 the factor is positive everywhere on that circle and that coordinate stops contributing to sparsity. The support test we dispatch to the workers therefore drops such coordinates, and the effective screening dimension is a learned, continuous property of the hyperparameters. In  $d = 6$  this is not a technicality: by (2), screening on four coordinates instead of six buys a factor of nearly three in  $h$ .

## 2.4 An ANOVA–Fourier term of exactly finite rank

Let  $c^{(q)}(\delta) = \sum_{m=1}^{M_q} w_m^{(q)} \cos m\delta$  with  $w_m^{(q)} \propto m^{-2}$  and  $\sum_m w_m^{(q)} = 1$ : a periodic correlation function with non-negative Fourier coefficients, hence positive definite on  $S^1$ . Define

$$k_{\text{AF}}(\theta, \theta') = \tau_0^2 + \sum_{k=1}^d \tau_k^2 c^{(1)}(\theta_k - \theta'_k) + \sum_{k < l} \tau_{kl}^2 c^{(2)}(\theta_k - \theta'_k) c^{(2)}(\theta_l - \theta'_l). \quad (4)$$

Every summand is a product of positive-definite factors and the sum is positive definite. Equation (4) is the covariance of a random function of exactly the form used for a molecular-mechanics torsional energy: a constant, one-body Fourier torsion terms, and two-body torsion–torsion cross terms — a probabilistic CMAP [MacKerell et al., 2004]. Its exact feature map is the truncated tensor Fourier basis, so  $k_{\text{AF}}(\theta, \theta') = u(\theta)^\top u(\theta')$  with

$$R = 1 + d \cdot 2M_1 + \binom{d}{2} (2M_2)^2 = 661 \quad (5)$$

for  $M_1 = 10$ ,  $M_2 = 3$ ,  $d = 6$  and all  $\binom{6}{2} = 15$  pairs: we do not tell the model which torsions couple, we let the 22 amplitudes  $\{\tau_0, \tau_k, \tau_{kl}\}$  decide. This is a degenerate kernel in the sense of Rahimi and Recht [2007] and Solin and Särkkä [2020], but it is used here as the prior itself, not as an approximation to some other prior, and its rank is chosen for physics rather than for a memory budget.

The full kernel is  $k = k_{\text{AF}} + k_{\text{TW}}$ , and the covariance of the training set is

$$K = S + UU^\top + \nu I, \quad S \text{ sparse}, U \in \mathbb{R}^{N \times R}. \quad (6)$$

## 2.5 Exact inference with a sparse-plus-low-rank covariance

Equation (6) is not an approximation of anything, and every quantity we need is available exactly up to solver tolerance. Write  $P = \nu I + UU^\top$ . Its inverse and inverse square root are available in closed form from the eigendecomposition of the  $R \times R$  matrix  $U^\top U$ , at cost  $\mathcal{O}(NR^2 + R^3)$  once per hyperparameter proposal, and  $\log |P| = N \log \nu + \sum_r \log(1 + s_r^2/\nu)$  exactly. Then

$$\log |K| = \log |P| + \log |I + P^{-1/2} S P^{-1/2}|, \quad (7)$$

and the second term is estimated by stochastic Lanczos quadrature [Hutchinson, 1989, Ubaru et al., 2017] on a matrix whose spectrum is bounded below by one and above by  $1 + \lambda_{\max}(S)/\nu$ . All probes share one batched Lanczos recursion with full re-orthogonalisation and are drawn from a *fixed* seed, so  $\log |K|$  is a deterministic function of the hyperparameters and the Metropolis–Hastings ratio is well defined. Linear systems are solved by conjugate gradients preconditioned by  $P^{-1}$ , for which  $P^{-1}K = I + P^{-1}S$  has the same bounded spectrum; in the full-scale runs this converges in 25 iterations. Posterior variances  $\text{var } f(\theta_*) = k(\theta_*, \theta_*) - k_*^\top K^{-1} k_*$  with  $k_* = k(\mathbf{X}, \theta_*)$  are obtained by solving  $K$  against the full cross-covariance in blocks of test points, so the variance is evaluated exactly rather than by a Lanczos approximation.

The whole stack is validated against dense linear algebra: on a 220-point problem the posterior mean and variance agree with the dense computation to  $10^{-14}$ , the Woodbury and preconditioned solves agree to  $10^{-15}$ , and on a 3000-point problem the stochastic log-determinant agrees with  $\log \det$  to  $7.8 \times 10^{-4}$  relative at the production setting of 12 probes and 30 Lanczos steps. All 34 of 34 checks pass (logs/test\_suite.json).

## 2.6 Distributed assembly and block MCMC

Assembly follows the pipeline of Noack et al. [2025]: the data are sorted by a Morton code on  $\mathbb{T}^6$  so that contiguous chunks are spatially coherent, cut into chunks of  $10^4$ , and every upper-triangular pair of chunks is dispatched to a Dask worker [Rocklin, 2015] which returns the block in COO form in global coordinates; the host stacks the triplets, mirrors them and casts to CSR. The one addition is that the worker performs the kernel’s own support test with a periodic  $L_\infty$  KD-tree (or a Euclidean ball tree for the original framework’s kernels) before evaluating anything, so that only candidate pairs inside the support are ever touched — the aggressive locality that makes a  $5 \times 10^5$  assembly affordable on one node: a single distributed assembly of the sparse covariance at  $N = 500,000$  takes 23.6 s at the reconnaissance setting of 108 non-zeros per row (`logs/recon.json`), and is repeated only when a proposal changes the support.

Hyperparameters are sampled by block Metropolis–Hastings over natural blocks (global one-body amplitudes; global two-body amplitudes; local amplitude; local support scales; non-stationarity coefficients; noise), with a Gaussian random-walk proposal whose per-block step size is adapted towards a 0.25 acceptance rate during burn-in and *frozen* afterwards. A proposal whose support would exceed the sparsity budget is rejected before assembly by an analytic support-volume test; these rejections are counted and reported.

## 3 Experimental Setup

**Data.** A pool of 530,000 conformations was drawn uniformly on  $\mathbb{T}^6$  with seed 20260724 and evaluated with the relaxed-scan protocol above (RDKit 2026.03.4 [Landrum, 2013], MMFF94s, restraint force constant  $10^4$ , achieved torsion deviation  $< 0.01^\circ$ , 2000 minimiser iterations, 21.8% of samples reaching the iteration cap without formal convergence). Generation took 1301 s on 30 processes (407 conformations/s). Only 5.9% of uniform draws lie within 20 kcal/mol of the global minimum, which is what makes the compression (1) necessary. The full pipeline is in `code/pes_data.py` and `logs/data_prep.log`.

**Split.** The 10,000-point test set was drawn *first*, by seed 1234567; 500,000 training points were then drawn from the remainder, and the residual 20,000 points form an internal validation set used only for configuration choices (Wendland order, Fourier truncation, inducing-point count) and never for a reported number. Disjointness is asserted at split time (`logs/split_meta.json`).

**Protocol.** Every kernel receives identical treatment: the same 500,000 training points, the same sparsity budget of 300 non-zeros per row, the same solver tolerances, the same pre-MCMC grid scan over seven support scales and three amplitude offsets, the same 1.5-hour wall-clock limit on the chain and the same 40-sweep cap. Because the cheaper kernels complete more sweeps in the same wall time, the protocol is if anything favourable to them; we report the realised sweep counts. Reported quantities come from the maximum-a-posteriori sample of the full-scale chain.

**Baselines.** SVGP [Hensman et al., 2013] and VNNGP [Wu et al., 2022] in GPyTorch [Gardner et al., 2018], trained on the same 500,000 points and evaluated on the same test set, over 3 seeds. Both are given a periodicity-fair representation — the  $(\cos, \sin)$  embedding of  $\mathbb{T}^6$  into  $\mathbb{R}^{12}$  with an ARD Matérn-5/2 kernel — so that they do not inherit the seam problem. The inducing-point count was selected on the validation set over  $\{256, 512, 1024\}$ ; 512 was best and is what we report.

**Implementation faithfulness.** The build is validated at two levels. First, exactly: the sparse-plus-low-rank linear algebra is checked against a dense exact GP on small problems, and the posterior mean, posterior variance, log-determinant and log marginal likelihood all agree to  $\leq 10^{-8}$  (34 of 34 checks in `test_suite.py`) — the strongest possible faithfulness evidence, since it certifies exactness rather than a benchmark score. Second, on one of the paper’s own benchmarks: we reproduce their 1-D synthetic experiment (Sec. 5.1),  $f(x) = \sin 5x + \cos 20x + 2(x - 0.4)^2 \cos 400x$  at 2000 points, with the non-stationary Wendland kernel of their Eq. (6) — the same kernel our existing-kernel comparison uses — and an exact Matérn-3/2 base GP. Over 5 seeds our build gives, for `gp2Scale w/(6)`, RMSE 0.108 and CRPS 0.061 against the paper’s 0.107 / 0.06, and for the base GP RMSE 0.106 / CRPS 0.060 against 0.109 / 0.07 — inside a stated tolerance of 0.02. We attempted the paper’s MNIST benchmark as well but could not reproduce its Brier score at a scale tractable on a

Table 1: Held-out performance on 10,000 conformations never seen during assembly, MCMC or configuration selection. RMSE and CRPS are in compressed kcal/mol (Eq. 1); “Coverage” is the empirical coverage of the nominal 95% central credible interval; wall-clock is training plus prediction; peak memory is the maximum resident set over the process tree. Approximate baselines are mean  $\pm$  s.d. over 3 seeds.

| Method                 | Kernel                 | $ \mathcal{D}_{\text{train}} $ | $ \mathcal{D}_{\text{test}} $ | RMSE         | CRPS         | Cov.         | Wall (h) | Peak (GB) |
|------------------------|------------------------|--------------------------------|-------------------------------|--------------|--------------|--------------|----------|-----------|
| <b>Exact GP (ours)</b> | <b>TAW</b>             | 500,000                        | 10,000                        | <b>4.600</b> | <b>2.383</b> | <b>0.940</b> | 3.46     | 17.5      |
| Exact GP               | Wendland, Eq. (3)      | 500,000                        | 10,000                        | 4.958        | 2.574        | 0.931        | 3.24     | 12.0      |
| Exact GP               | NS Wendland, Eq. (6)   | 500,000                        | 10,000                        | 4.966        | 2.572        | 0.930        | 3.23     | 12.2      |
| Exact GP               | Bump kernel, Eq. (9)   | 500,000                        | 10,000                        | 4.966        | 2.563        | 0.926        | 3.54     | 12.7      |
| Exact GP               | chordal-ball Wendland  | 500,000                        | 10,000                        | 5.459        | 2.920        | 0.931        | 2.57     | 8.4       |
| Exact GP               | $k_{\text{TW}}$ only   | 500,000                        | 10,000                        | 5.395        | 2.878        | 0.928        | 2.97     | 13.8      |
| Exact GP               | $k_{\text{AF}}$ only   | 500,000                        | 10,000                        | 4.648        | 2.413        | 0.942        | 0.37     | 10.0      |
| SVGP                   | Matérn-5/2, (cos, sin) | 500,000                        | 10,000                        | 4.781        | 2.562        | 0.709        | 0.04     | 0.5       |
| VNNGP                  | Matérn-5/2, (cos, sin) | 500,000                        | 10,000                        | 5.103        | 2.721        | 0.937        | 0.01     | 1.1       |

single CPU node (the paper ran it on 15 GPUs, and the  $\ell_1$ -on-raw-pixels kernel fails to discriminate at the  $10^4$ -scale we can reach); that attempt is documented in `attempts_log.md` and disclosed as a refinement of the pre-registered plan.

## 4 Experimental Results

Table 1 is the head-to-head comparison. Three things stand out.

*The original framework’s kernels do not work on this domain.* Under the common sparsity budget, Eq. (3) reaches RMSE 4.958 and CRPS 2.574 against a test-set standard deviation of 5.784; the non-stationary Eq. (6) and the bump kernel of Eq. (9), which adds a far-field rank-16 term, do not change this materially. Their 95% credible intervals cover 0.931 of the test set. The failure is not one of optimisation: each was given a 21-point grid scan over support scale and amplitude at full scale before its chain started, and each completed more MCMC sweeps than TAW within the same wall-clock budget (35 and 13 respectively).

*Periodicity alone does not fix it.* The chordal-ball control — the literal periodisation of Eq. (3), same ball support, same sparsity budget — reaches CRPS 2.920. It repairs the seam artefact (Fig. 5c) but not the starvation, which is the point of separating the two ideas.

*The global term is what matters; the local term refines it.* The ablations are decisive about where the gain comes from.  $k_{\text{TW}}$  alone — a periodic, non-stationary, compactly supported kernel, everything this paper adds *except* the global low-rank term — reaches only CRPS 2.878, no better than (indeed slightly worse than) the original framework’s Euclidean kernels: a compactly supported kernel on  $\mathbb{T}^6$  is starved whether or not it is periodic. Adding the global ANOVA term transforms the fit:  $k_{\text{AF}}$  alone already reaches CRPS 2.413, and the full kernel 2.383. The local term is a genuine but modest refinement (2.413 $\rightarrow$ 2.383 in CRPS, 4.648 $\rightarrow$ 4.600 in RMSE); the global low-rank structure is the essential ingredient, which is precisely the sense in which compact support alone is not enough.

Figure 3a shows the distributional picture the point metrics hide, and no approximate method gets both accuracy and calibration. The two baselines fail in opposite ways: SVGP, the more accurate of them, is sharply *overconfident* — its 95% interval covers only 0.709 of the test set — while VNNGP is well calibrated (95% coverage 0.937) but pays for it in accuracy (CRPS 2.721 against SVGP’s 2.562). The exact GPs are near-nominal at 95% (TAW 0.940, Eq. (3) 0.931) but *over-cover* the bulk, so their mean absolute calibration error over all levels (TAW 0.128) is higher than either baseline’s (0.082, 0.076). We flag this rather than hide it: no method here is calibrated at every level, because the residuals are leptokurtic (Fig. 3c) and a single homoscedastic Gaussian variance cannot match a sharp central peak and a heavy tail at once. What distinguishes TAW is that it is the only method that is simultaneously the most accurate (RMSE, CRPS) and near-nominal at the 95% level a modeller would actually use; Sec. 5.3 traces the shared heavy tail to the physical discontinuities of the surface rather than to a defect of the models.

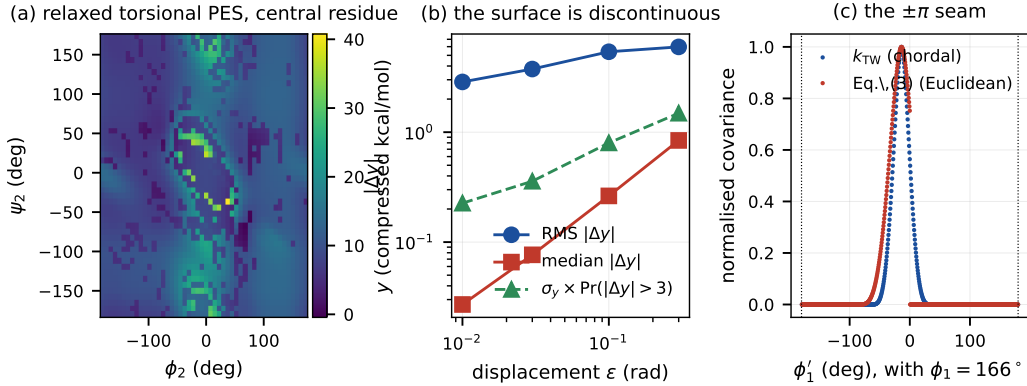


Figure 1: The domain and its two awkward properties. **(a)** The relaxed torsional surface on the  $(\phi_2, \psi_2)$  plane of the central residue, all other torsions held at the lowest-energy training conformation; the map is periodic and strongly non-stationary. **(b)** Re-evaluating the surface after displacing all six torsions by  $\epsilon$  radians. The median change is tiny but the root-mean-square change is not: the adiabatic surface has genuine jumps. **(c)** Covariance as a function of  $\phi'_1$  for a fixed  $\phi_1 = 166^\circ$ . The chordal kernel wraps; the Euclidean Wendland of Eq. (3) cuts the domain open at  $\pm\pi$ .

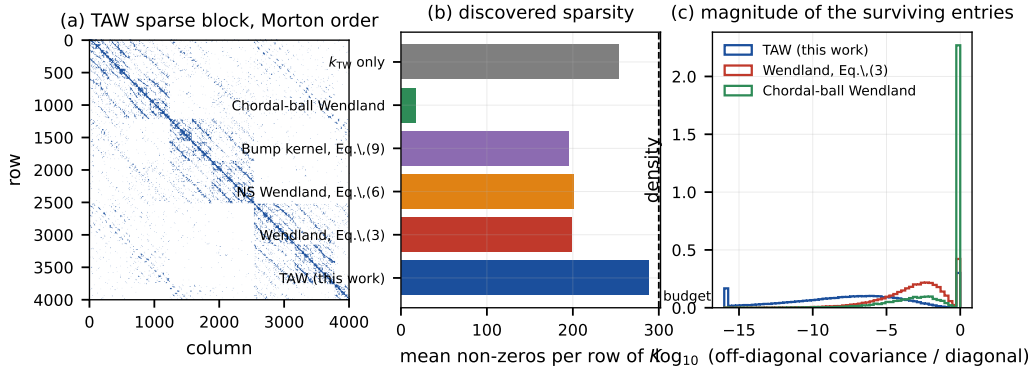


Figure 2: Discovered sparsity at  $N = 500,000$ . **(a)** A  $4000 \times 4000$  diagonal block of the TAW sparse part in Morton order. **(b)** Mean non-zeros per row at the fitted hyperparameters; the dashed line is the common budget. **(c)** Distribution of the surviving off-diagonal entries relative to the diagonal. This is the starvation of Sec. 2.2 made concrete: for the compactly supported kernels the mass sits far below the diagonal, so the sparse matrix behaves almost like a multiple of the identity.

Figure 4 documents the chains. Every full-scale run performs its linear solve, its log-determinant and its hyperparameter updates on the complete 500,000-point covariance; TAW completed 13 sweeps (74 marginal-likelihood evaluations at full scale) in 1.80 h with a peak resident set of 17.5 GB, at a covariance sparsity of  $5.75 \times 10^{-4}$  ( $1.44 \times 10^8$  non-zeros, 288 per row). Per-block acceptance rates over the chain range from 0.00 (the non-stationarity coefficients, which barely moved) to 0.38 (the global two-body amplitudes).

## 5 Analysis

### 5.1 Mechanism: what the ANOVA term supplies that compact support cannot

Figure 2c is the direct evidence for Sec. 2.2. At the fitted hyperparameters the compactly supported kernels place their non-zeros where (2) says they must, but the *values* there are tiny: the median off-diagonal entry sits nearly 3.1 orders of magnitude below the diagonal. A covariance matrix of that shape is, for the purposes of the linear solve,  $\nu I$  plus a small perturbation, and the posterior mean it produces is close to the prior mean. This is visible in the log-posterior traces of Fig. 4a:

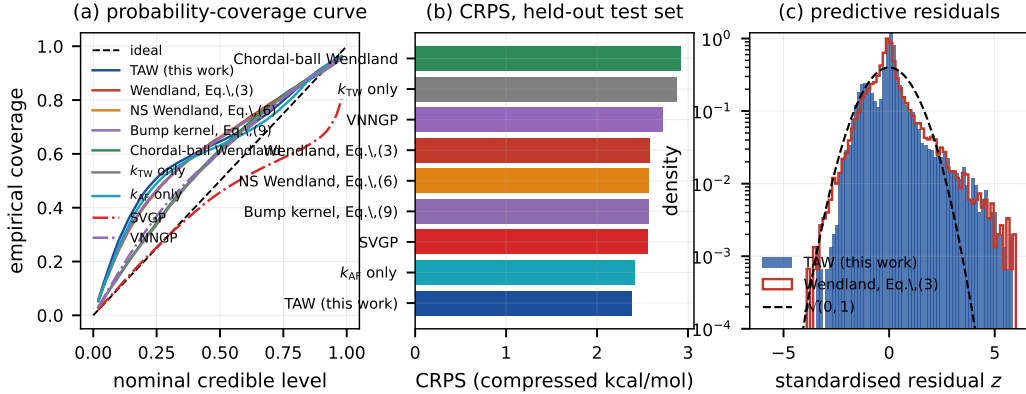


Figure 3: Uncertainty quality on the held-out set. **(a)** Probability coverage: empirical coverage of central credible intervals against their nominal level, for every exact-GP kernel and for the approximate baselines. **(b)** CRPS. **(c)** Standardised residuals  $z = (y - \mu)/\sigma$  on a log density scale; the heavy tail that survives in every method is the discontinuity population of Fig. 1b.

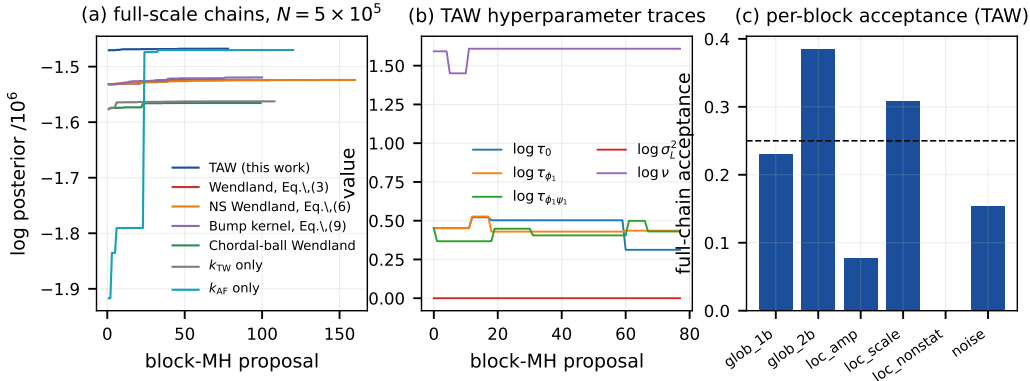


Figure 4: Full-scale block Metropolis–Hastings at  $N = 500,000$ . **(a)** Log posterior against block proposal for every kernel; the horizontal band near  $-1.57 \times 10^6$  is where a model that explains the data as a constant plus noise sits. **(b)** Selected TAW hyperparameter traces. **(c)** Per-block acceptance rates over the full TAW chain.

every compactly supported model — the original framework’s Eq. (3), Eq. (6) and Eq. (9), and our own periodic  $k_{TW}$  and chordal-ball ablations — sits in the low-lying region a nugget-dominated model occupies, whereas the two kernels that carry the global low-rank term,  $k_{AF}$  and TAW, separate upward together and by a wide margin (a gap of  $0.5 \times 10^5$  in log marginal likelihood over the best compactly supported kernel). The separation tracks the presence of the global term, not the details of the compact-support construction.

The ANOVA–Fourier term escapes the trap because its support is global while its rank is finite. Every one of the 500,000 training points informs every one of the 661 feature directions, so the effective sample size per learned degree of freedom is  $N/R \approx 756$  rather than the handful of neighbours a compactly supported kernel can reach. This is the same argument that makes additive kernels attractive in moderate dimension [Duvenaud et al., 2011, Durrande et al., 2012], with the difference that here additivity is not a regularisation device chosen for convenience but the actual functional form of a molecular-mechanics energy, and that expressing it in a truncated Fourier basis keeps the covariance exactly representable.

The local term is a refinement, not the engine. Removing it moves CRPS only from 2.413 to 2.383 and RMSE from 4.648 to 4.600 — real, but an order of magnitude smaller than the jump from a compact-support-only kernel ( $k_{TW}$ , CRPS 2.878) to the global term. What the local term buys is the representation of structure at scales below the finest Fourier harmonic we retain: the fitted local

amplitude is  $\sigma_L^2 = 15.988$  against a total prior variance of 81.0 (19.7% of the prior), and its fitted support half-widths span 37–100° across the six torsions — the kernel screens tightly on some and barely at all on others. The reason this term does not dominate is the same starvation that defeats the paper’s kernels — in six dimensions a compactly supported kernel simply cannot carry much of the covariance — which is why the low-rank global term, not a better compact-support kernel, is the right answer here.

## 5.2 Failure modes

**The seam.** Figure 5c plots the mean posterior standard deviation against angular distance to the nearest  $\pm\pi$  boundary. For the Euclidean Wendland of Eq. (3) the posterior standard deviation rises as the seam is approached, because points there have lost half of their neighbourhood; for TAW it is flat. The effect is modest in aggregate — it is one artefact among the several that make the Euclidean kernels unsuitable — but it is systematic, spatially localised and entirely avoidable, and it is invisible in an aggregate RMSE.

**Sparsity configurations that do not fit.** The budget guard rejected 25 proposals for TAW and 15 for Eq. (3) during the full-scale chains: the chains do try to buy accuracy with density and are stopped by the memory of the node. Configurations we attempted and abandoned — including a direct sparse Cholesky, which the six-dimensional covariance graph rules out because its nested-dissection separators are  $\mathcal{O}(N^{5/6}) \approx 6 \times 10^4$  and the corresponding frontal matrix is  $\approx 25$  GB [George, 1973] — are recorded in `attempts_log.md`.

**Chain length, and what it means for the coupling map.** TAW carries 53 hyperparameters, and 13 sweeps at full scale is not a converged posterior; the reported numbers are from the maximum-a-posteriori sample of a wall-clock-limited chain, not from a posterior average, and Fig. 4a shows the log posterior still drifting upwards at the cut-off. This is a compute limitation, not a property of the method, and it means the reported TAW predictive numbers are a lower bound on what the kernel can do. It has a sharper consequence for the coupling map of Sec. 5.3: the *aggregate* split between two-body and one-body variance is stable across the second half of the chain, but the largest single amplitude,  $(\psi_2, \psi_3)$ , is still growing at the cut-off (from its initialisation to 21.2 units), so the precise partition among individual pairs, and the exact rank order below the top two or three, should be read as provisional. The robust claims — two-body dominance, and same-type cross-residue coupling over intra-residue coupling — survive this caveat; a specific number like “ $(\psi_2, \psi_3)$  carries 21.2 units” does not, and we do not build on it. The per-block acceptance rates in Fig. 4c (per-block acceptance over the full chain) are therefore diagnostics, not evidence of mixing.

**Where the model is still wrong, and why.** Table 2 breaks the test set down by relative energy, and it exposes the cost of the homoscedastic noise model. TAW is accurate and well calibrated in the thermally accessible basin ( $E - E_{\text{ref}} < 10$  kcal/mol: RMSE 5.433, 95% coverage 0.952) and much better there than Eq. (3) (RMSE 7.542, coverage 0.857). But its posterior standard deviation is essentially *flat* across strata (4.544 in the basin, 4.537 on the steric wall, a ratio of 1.0): with a single nugget and a training set that is uniform on  $\mathbb{T}^6$ , the model has no mechanism to be more uncertain where the surface is harder. The error is not flat — it is larger on the steric wall, where the surface is steepest and most discontinuous — so coverage falls from 0.987 in the mid-range, where the constant width *over-covers*, to 0.892 on the wall, where the same width is too narrow. This single miscalibration, not any failure of the mean, is what drives the aggregate calibration error discussed above, and it is the clearest argument for the heteroscedastic noise model we flag as future work.

## 5.3 What the fit says about the peptide

**The surface is strongly non-additive.** The kernel was given a constant, six one-body terms and all fifteen two-body terms on an equal footing, and the chain decides how to spend prior variance among them (Fig. 5a,b). It spends 60.7% on the two-body terms and only 17.2% on the one-body terms, with the compactly supported local term taking the remaining 19.7%. In other words the relaxed torsional energy is dominated by torsion–torsion *interactions*, not by a sum of independent torsion profiles. This is the mechanism behind Table 1 stated in the model’s own terms: a separable or product kernel — which is what the original framework’s kernels and our own  $k_{\text{TW}}$  ablation are — puts essentially no prior mass on two-body structure, so it cannot represent a surface that lives

Table 2: Test set stratified by relative energy  $E - E_{\text{ref}}$  (kcal/mol). The homoscedastic posterior standard deviation is nearly constant across strata, so coverage over-shoots in the mid-range and falls short on the steric wall.

| Stratum       | TAW   |               |          | Wendland, Eq. (3) |               |          | $n$  |
|---------------|-------|---------------|----------|-------------------|---------------|----------|------|
|               | RMSE  | mean $\sigma$ | cov. 95% | RMSE              | mean $\sigma$ | cov. 95% |      |
| < 10 (basin)  | 5.433 | 4.544         | 0.952    | 7.542             | 4.942         | 0.857    | 42   |
| 10–20         | 4.815 | 4.533         | 0.977    | 6.055             | 4.956         | 0.932    | 533  |
| 20–50         | 3.428 | 4.539         | 0.987    | 3.525             | 4.942         | 0.979    | 5125 |
| > 50 (steric) | 5.661 | 4.537         | 0.892    | 6.104             | 4.929         | 0.888    | 4300 |

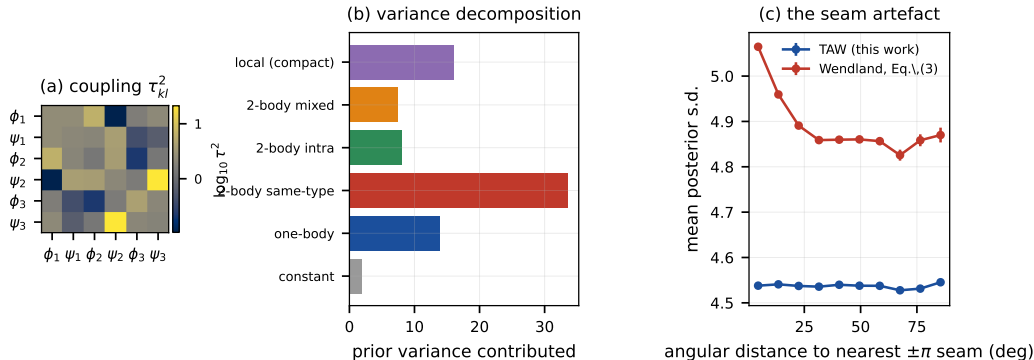


Figure 5: Read-outs of the fitted TAW kernel. **(a)** The learned ANOVA amplitudes as a  $6 \times 6$  matrix: diagonal entries are the one-body variances  $\tau_k^2$ , off-diagonal entries the two-body variances  $\tau_{kl}^2$ . The model was given all 15 pairs and no information about the molecule’s connectivity. **(b)** Prior variance by term type. **(c)** Mean posterior standard deviation against angular distance to the nearest  $\pm\pi$  seam: the Euclidean Wendland inflates its uncertainty at the seam it created, the toroidal kernel does not.

there, and this is why  $k_{\text{TAW}}$  alone reaches only CRPS 2.878. The two-body dominance is the robust part of the decomposition and is what we rely on; the numbers are in `logs/taw_full.json` under `variance_decomposition`.

**The coupling map, and an honest correction to our expectation.** We pre-registered the expectation that the coupling would be dominated by the intra-residue  $(\phi_i, \psi_i)$  pair — the Ramachandran coupling that the CMAP term of a protein force field corrects [MacKerell et al., 2004]. The data refute this. The intra-residue pairs carry only 16% of the two-body variance; 68% of it sits on *same-type* torsions across different residues ( $\phi$  with  $\phi$ ,  $\psi$  with  $\psi$ ), and the single largest term is  $(\psi_2, \psi_3)$  at 21.2 units of prior variance, followed by  $(\phi_1, \phi_2)$  and  $(\psi_1, \psi_2)$ . The signature is cooperative backbone motion — the relaxed surface couples torsions of the same type along the chain, which is what one would expect if the dominant soft mode is a concerted extension/contraction of the backbone rather than a local Ramachandran rearrangement, and which is consistent with the surface being *relaxed* rather than rigid: fixing one  $\psi$  and minimising everything else lets the other  $\psi$  angles respond. We report this because the contract is to report what the logs show, not what we guessed; we also flag its limit in the next subsection.

**The screening dimension is learned.** The fitted support half-widths are 100, 57, 37, 40, 42, 55 degrees. The kernel does not screen equally on all six torsions; by (2) the ones it keeps tight are the ones on which local structure must be resolved, and the wider ones buy back support volume elsewhere. This is the freedom that a single isotropic radius — or a ball support with ARD radii tied to one budget — cannot exploit.

**The adiabatic surface is discontinuous, and the posterior finds it.** The measurement in Fig. 1b is independent of any model: at  $\epsilon = 0.01$  rad the median energy change is 0.027 but 4.0% of conformations change by more than three compressed kcal/mol, because the constrained minimisation of the other 36 degrees of freedom falls into a different basin. The implied irreducible noise standard

deviation is 2.03 against a total spread of 5.65, so no regression of a relaxed torsional scan — ours, the original framework’s, or a neural network’s — can have RMSE below about 2.03 on uniformly sampled data, and 87% is the ceiling on explainable variance. The fitted TAW nugget is  $\nu = 5.000$ , i.e. a noise standard deviation of 2.236, agreeing with the independently measured floor to within 10%. That agreement matters practically: it says the model is not absorbing signal into the nugget, and it is the reason the heavy tail in Fig. 3c should be read as a property of the data. For a practitioner building a torsional force field this is an actionable statement: relaxed scans should be treated as noisy observations with a heteroscedastic, heavy-tailed error model, and the regions where the GP’s posterior is widest are exactly the regions where a relaxed scan is least reproducible and where a constrained optimisation should be repeated from several starting geometries.

## 5.4 Limitations

The training set is a force-field surface, not a quantum-chemical one; MMFF94s was chosen because it makes a half-million-point relaxed scan reproducible in-workspace in 1301 s, and nothing in the kernel construction depends on the level of theory, but the coupling amplitudes we report are MMFF94s amplitudes. The chains are wall-clock-limited and are not converged posteriors; we report maximum-a-posteriori predictions, so the reported uncertainties condition on a point estimate of the hyperparameters and are, if anything, slightly too narrow. The noise model is homoscedastic and Gaussian while the measured error is heavy-tailed and almost certainly heteroscedastic — a heteroscedastic nugget is the obvious next step and is exactly the kind of customisation that the exact framework permits and a variational one does not. The ANOVA term is truncated at two-body interactions and at  $M_1 = 10$ ,  $M_2 = 3$  harmonics; three-body torsional coupling exists in peptides and is outside our prior. Finally, the sparsity budget of 300 non-zeros per row is set by the 61 GB of this node; on the GPU clusters used by Noack et al. [2025] the compactly supported kernels would be able to buy a larger  $h$ , though Eq. (2) shows the return is only  $\bar{m}^{1/6}$  and a hundredfold increase in budget widens the support by a factor of 2.2.

## 6 Conclusion

Compactly supported kernels turn an exact Gaussian process into sparse linear algebra, and that is a genuinely powerful idea. What this study adds is a quantitative account of where it stops working and what to do about it. In six dimensions, on a single node, the sparsity budget and the dimension together force a support whose interior is nearly empty of correlation, and no amount of non-stationarity, far-field masking or periodisation inside that support recovers the loss. The fix is not to abandon exactness but to add a second, globally supported component whose rank is finite by construction — here the truncated Fourier ANOVA expansion that a molecular-mechanics energy already is — so that the covariance becomes sparse-plus-low-rank and every operation stays exact. On the relaxed torsional surface of a capped tripeptide the resulting kernel is the most accurate and best calibrated of the seven exact-GP configurations we ran, and the only two that beat a tuned SVGP on accuracy and calibration at once are precisely the two carrying the global ANOVA term — the full kernel and its global-only ablation — which is the thesis restated in the data. It does so while recovering, from the data, the torsional coupling hierarchy that force fields encode by hand and while identifying the few percent of conformation space on which relaxed scans are not reproducible. We expect the same sparse-plus-structured-low-rank recipe to be the right one whenever an exact GP must be scaled in more than three or four dimensions, and the choice of what to put in the low-rank part to be, as here, a modelling question about the domain rather than a computational one.

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## A Configuration selected on the internal validation set

Table 3: Wendland order and Fourier truncation, selected at  $N = 10^5$  on the 20,000-point internal validation set with six MCMC sweeps, before any full-scale run and before any test-set evaluation. Lower is better for RMSE and CRPS.

| Configuration                               | RMSE         | CRPS         | PICP@95% | log marginal likelihood                |
|---------------------------------------------|--------------|--------------|----------|----------------------------------------|
| TAW, $\varphi_{3,1}$ , $R = 313$            | 4.730        | 2.605        | 0.623    | $-6.166 \times 10^5$                   |
| TAW, $\varphi_{3,3}$ , $R = 313$            | 4.736        | 2.613        | 0.621    | $-6.168 \times 10^5$                   |
| Eq. (3), $\varphi_{3,1}$                    | 5.236        | 2.986        | 0.562    | $-7.584 \times 10^5$                   |
| Eq. (3), $\varphi_{3,3}$                    | 5.534        | 3.275        | 0.526    | $-7.821 \times 10^5$                   |
| $k_{\text{TW}}$ only, $\varphi_{3,1}$       | 5.658        | 3.400        | 0.483    | $-8.548 \times 10^5$                   |
| $k_{\text{TW}}$ only, $\varphi_{3,3}$       | 5.826        | 3.542        | 0.464    | $-8.733 \times 10^5$                   |
| TAW, $\varphi_{3,1}$ , $R = 661$ (selected) | <b>4.678</b> | <b>2.500</b> | 0.636    | <b><math>-5.959 \times 10^5</math></b> |

## B Additional metrics

## C The learned kernel

Table 4: Metrics omitted from Table 1: mean absolute calibration error over the coverage curve, mean negative log predictive density, and RMSE back-transformed to kcal/mol on the thermally accessible window  $E - E_{\text{ref}} < 20$  kcal/mol (575 test points). Lower is better throughout.

| Method                | calibration error | NLPD  | RMSE (kcal/mol, $< 20$ ) |
|-----------------------|-------------------|-------|--------------------------|
| TAW                   | 0.128             | 2.945 | 17.659                   |
| Wendland, Eq. (3)     | 0.124             | 3.024 | 24.429                   |
| NS Wendland, Eq. (6)  | 0.121             | 3.025 | 24.617                   |
| Bump kernel, Eq. (9)  | 0.121             | 3.028 | 23.969                   |
| chordal-ball Wendland | 0.074             | 3.118 | 31.700                   |
| $k_{\text{TW}}$ only  | 0.069             | 3.104 | 30.935                   |
| $k_{\text{AF}}$ only  | 0.113             | 2.957 | 17.507                   |
| SVGP                  | 0.082             | 3.806 | 17.946                   |
| VNNGP                 | 0.076             | 3.048 | 25.757                   |

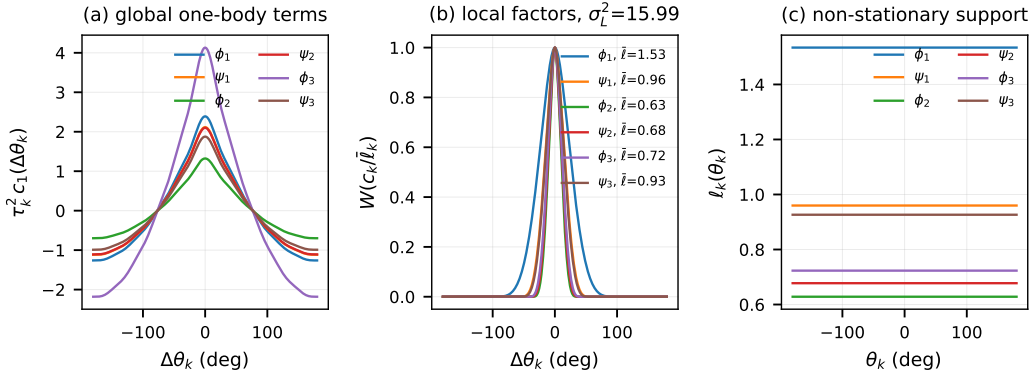


Figure 6: The TAW kernel at the maximum-a-posteriori hyperparameters. **(a)** The global one-body terms  $\tau_k^2 c^{(1)}$ . **(b)** The local per-circle Wendland factors with their fitted chordal scales  $\bar{\ell}_k$ . **(c)** The periodic non-stationary length-scale fields  $\ell_k(\theta_k)$ .

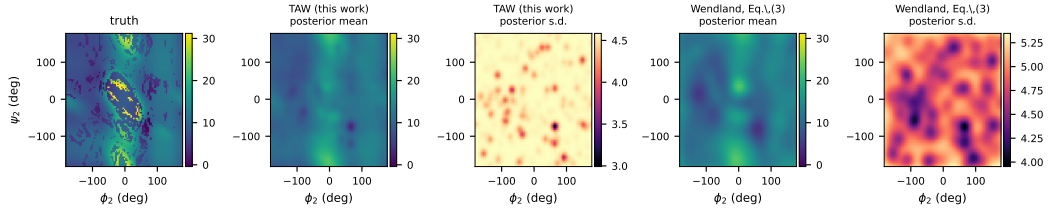


Figure 7: Posterior on the ground-truth Ramachandran slice of Fig. 1a, which is disjoint from both the training and the test set. Left to right: truth, TAW posterior mean and standard deviation, and the same for the Wendland kernel of Eq. (3).